Lecture 11 (10/9)

Autoregressive models, regression to the mean, and the Hurwicz bias

Unit 3 so far:

- An estimator $\hat{\theta}$ is computed from data and varies from replicate to replicate.
- Its sample distribution ($\approx$ Normal) is used to obtain the 95% confidence interval $\hat{\theta} \pm 1.96\,\mathrm{se}(\hat{\theta})$ and the p-value for $H_0 : \theta = 0$.
- Bias-variance decomposition: $\mathrm{MSE}_{\hat{\theta}} = \mathrm{se}(\hat{\theta})^2 + \mathrm{Bias}^2$.
- Least squares: $\hat{\beta}_0, \hat{\beta}_1$ minimize SSR; $\hat{\beta}_1 \approx \mathbf{Normal}\left(\beta_1, \sigma_\epsilon^2/(N\sigma_X^2)\right)$.

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Today:

- More practice working with (single-predictor) regression models
- Time series models.

p-values vs. R^2

Question: State whether each of the following scenarios is possible when we fit data to a linear regression model. The p-value (p_v) is for the null hypothesis $H_0 : \beta_1 = 0$; assume there are $N > 3$ data points.

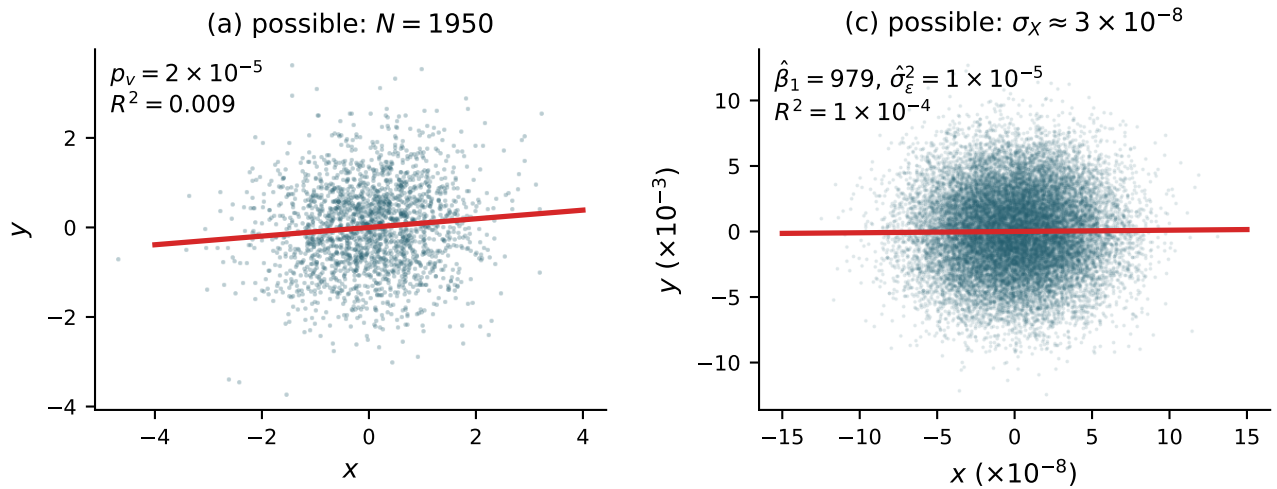
(a) $p_v = 10^{-5}$ and $R^2 = 10^{-2}$.

(b) $p_v = 0.5$ and $R^2 = 0.99$.

(c) $\hat{\beta}_1 = 1000, \hat{\sigma}_\epsilon^2 = 10^{-5}$ and $R^2 = 10^{-4}$.

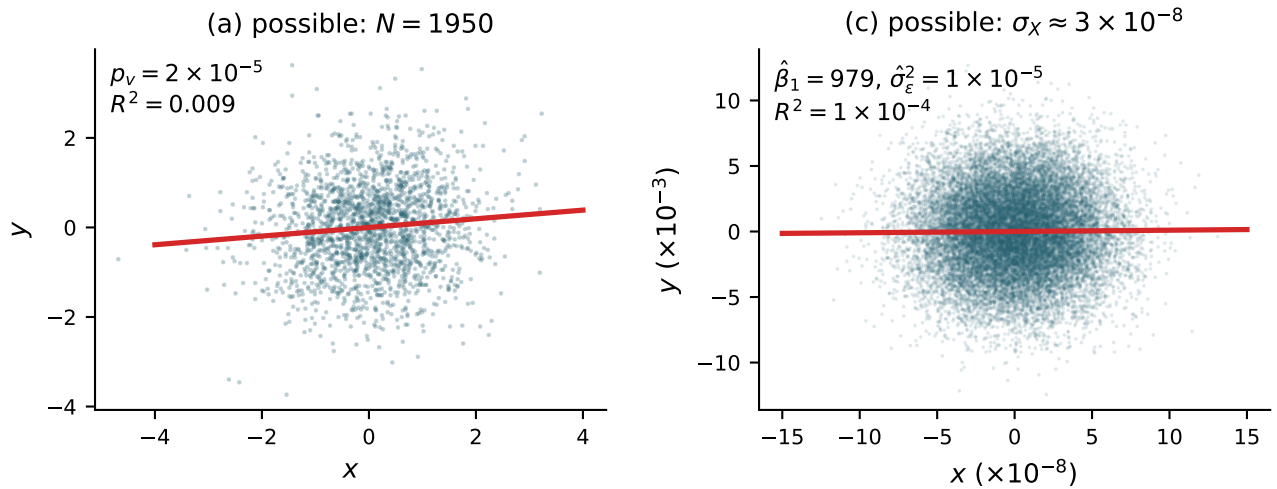
p-values vs. R^2: answers

(a) and (c) yes.



p-values vs. R^2: answers

(a) and (c) yes.



(b), no: for one predictor (see the drill "The test statistic and R^2" in Section 3.6 of the notes),

$$\boxed{\hat{T}^2 = (N-2)\,\frac{R^2}{1-R^2}}$$

so $R^2 = 0.99$ forces $\hat{T}^2 = 99(N-2)$ and a tiny p_V.

Estimating parameters in Python

With `x`, `y` numpy arrays. Using the covariance:

```
C = np.cov(x, y)
beta1_hat = C[0, 1] / C[0, 0]        # cov(x, y) / var(x)
```

Here `C` is the 2×2 sample covariance matrix

$$C = \begin{pmatrix} \widehat{\text{var}}(X) & \widehat{\text{cov}}(X, Y) \\ \widehat{\text{cov}}(X, Y) & \widehat{\text{var}}(Y) \end{pmatrix}, \qquad \widehat{\text{cov}}(X, Y) = \frac{1}{N-1} \sum_i (X_i - \overline{X})(Y_i - \overline{Y}),$$

so `C[0, 1] / C[0, 0]` $= \widehat{\text{cov}}(X, Y)/\widehat{\text{var}}(X) = \hat{\beta}_1$ (Python counts rows and columns from 0).

Estimating parameters in Python

We can also compute $\hat{\beta}_1$ directly from the formula:

```
beta1_hat = np.sum((x - np.mean(x)) * (y - np.mean(y))) \
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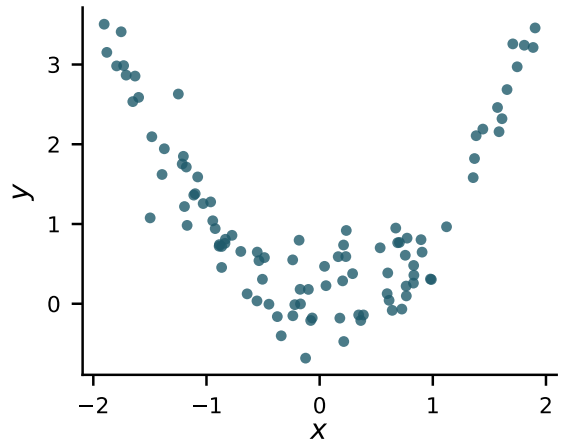
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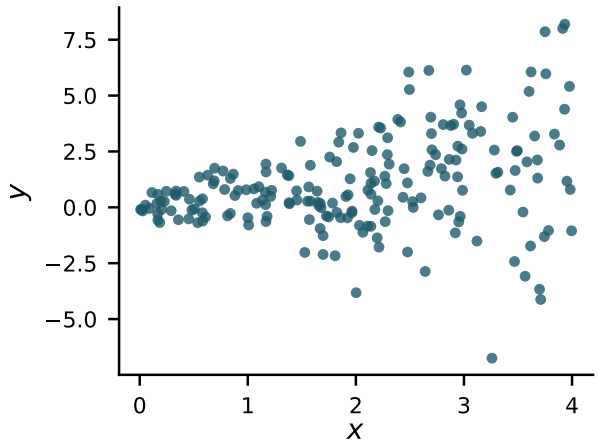
and the other estimators:

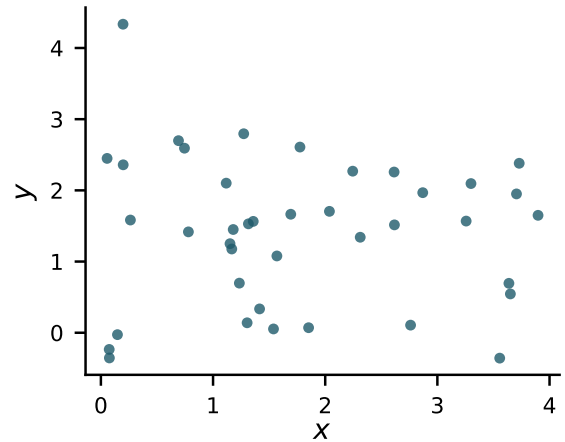
```
beta0_hat = np.mean(y) - beta1_hat * np.mean(x)
r = y - (beta0_hat + beta1_hat * x)           # residuals
sigma2_hat = np.sum(r**2) / (len(y) - 2)
R2 = 1 - np.sum(r**2) / np.sum((y - np.mean(y))**2)
se_beta1 = np.sqrt(sigma2_hat / np.sum((x - np.mean(x))**2))
T = beta1_hat / se_beta1
p_v = 2 * (1 - norm.cdf(np.abs(T)))           # from scipy.stats
```

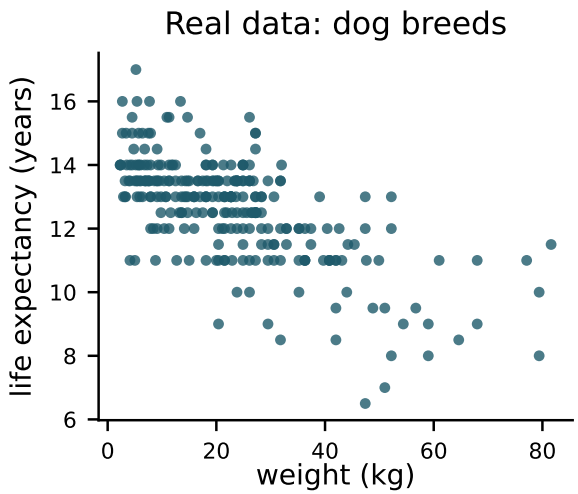
p-values vs. R^2

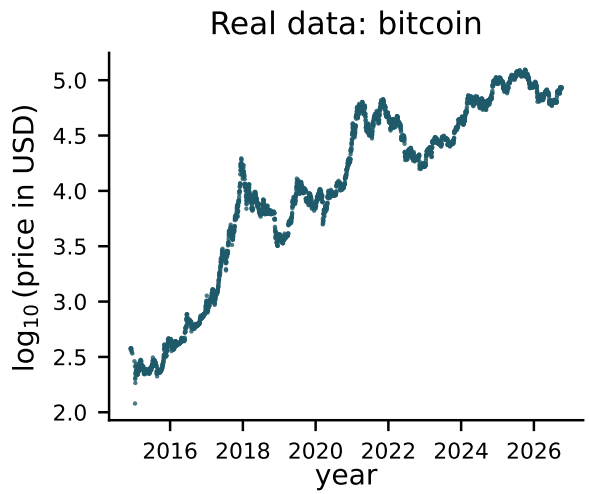
Guess R^2 and the p-value for $H_0 : \beta_1 = 0$ in each plot. No calculations needed!

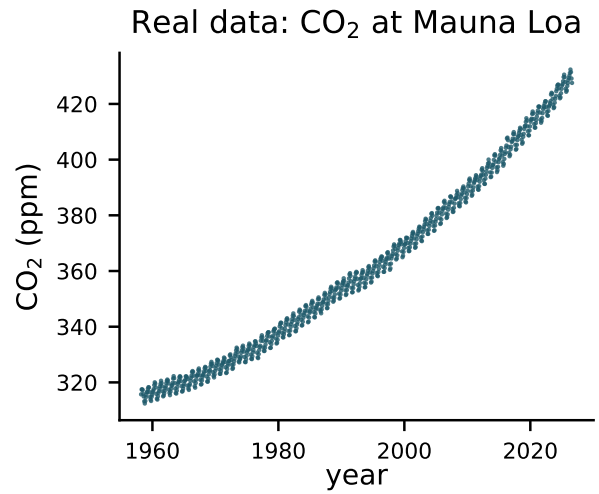






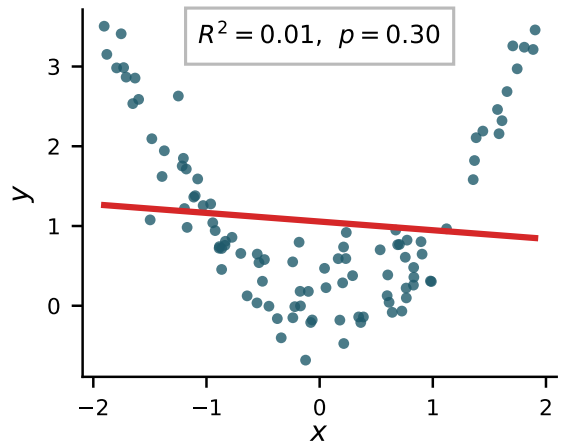


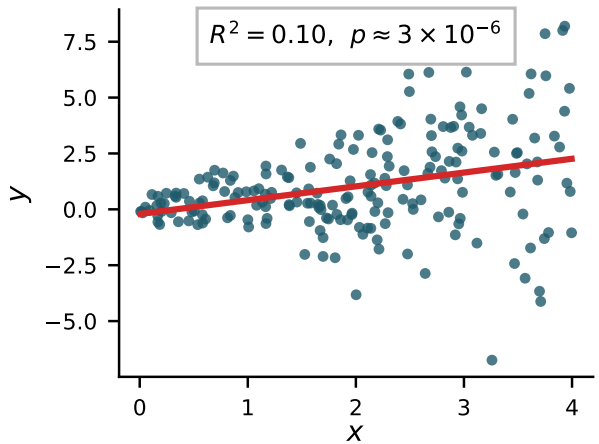


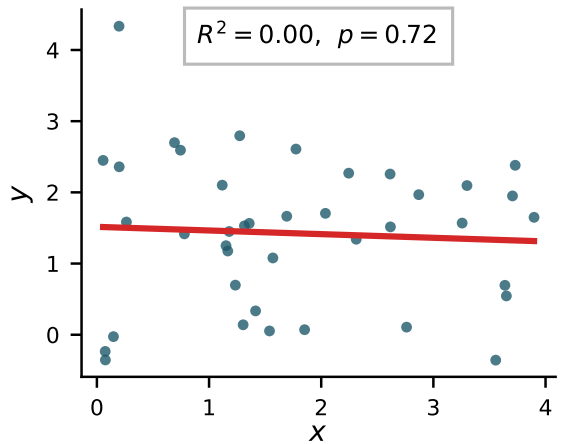


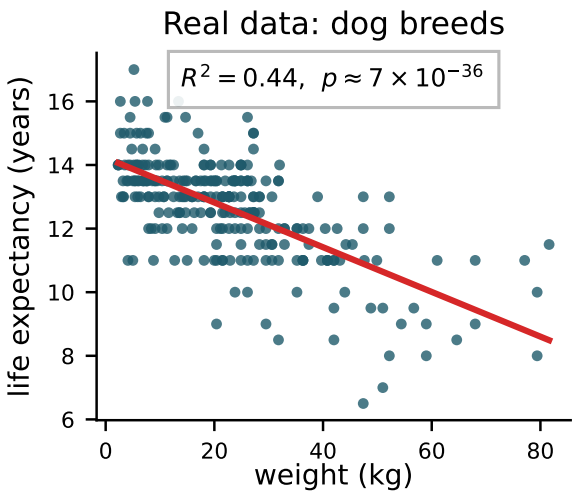
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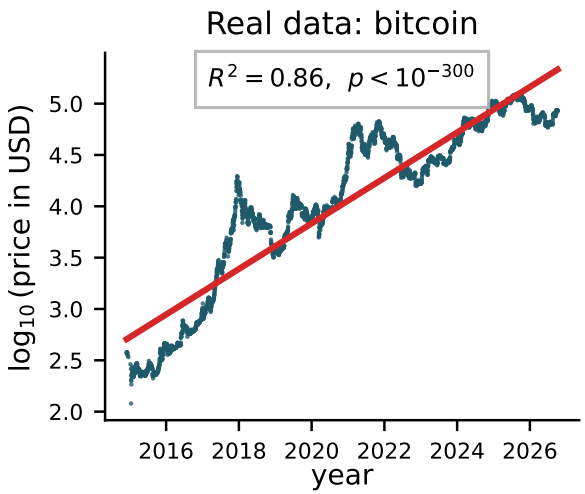
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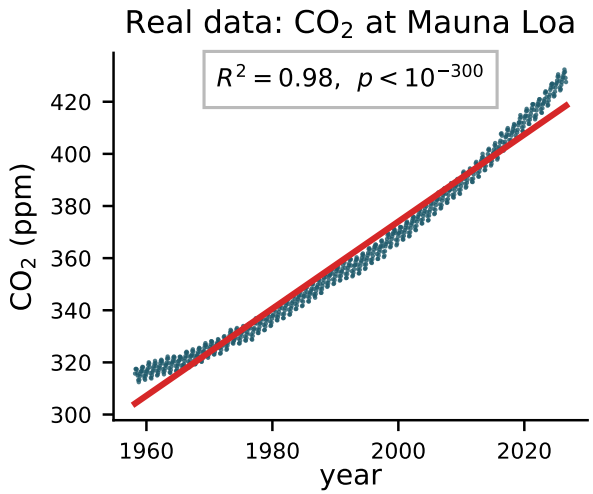






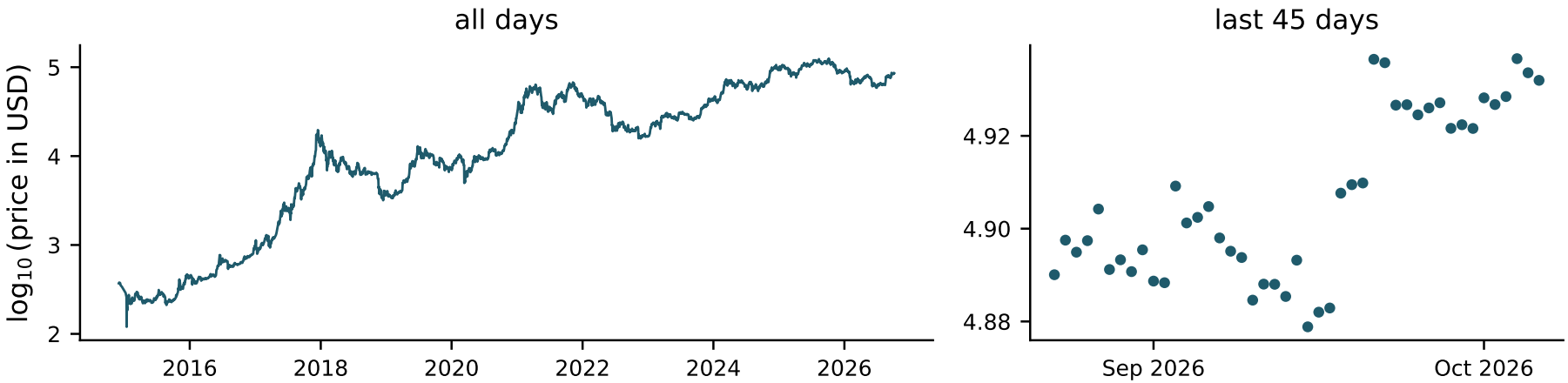






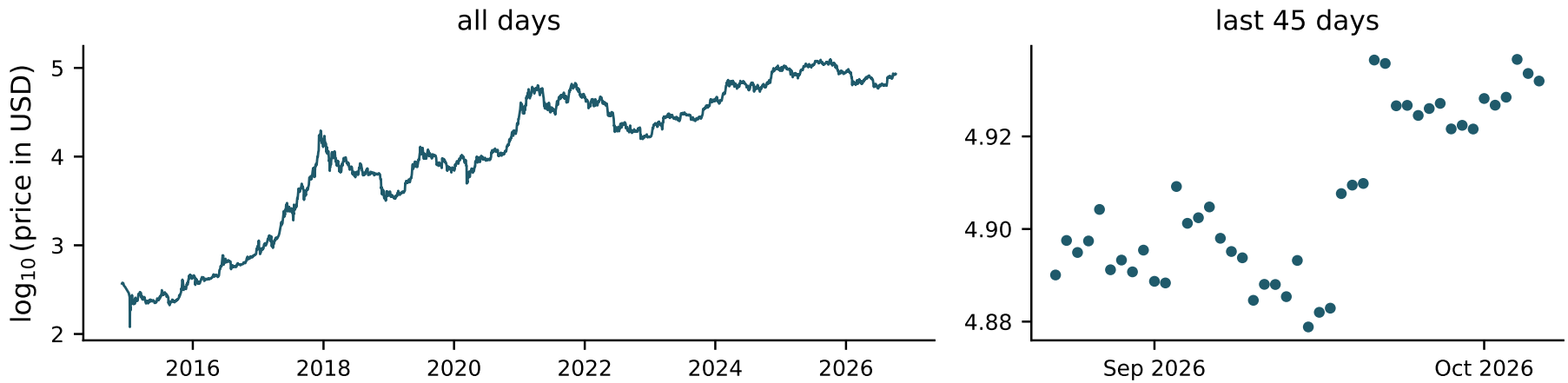
Time series data

Here $Y_t = \log(\text{price on day } t)$.



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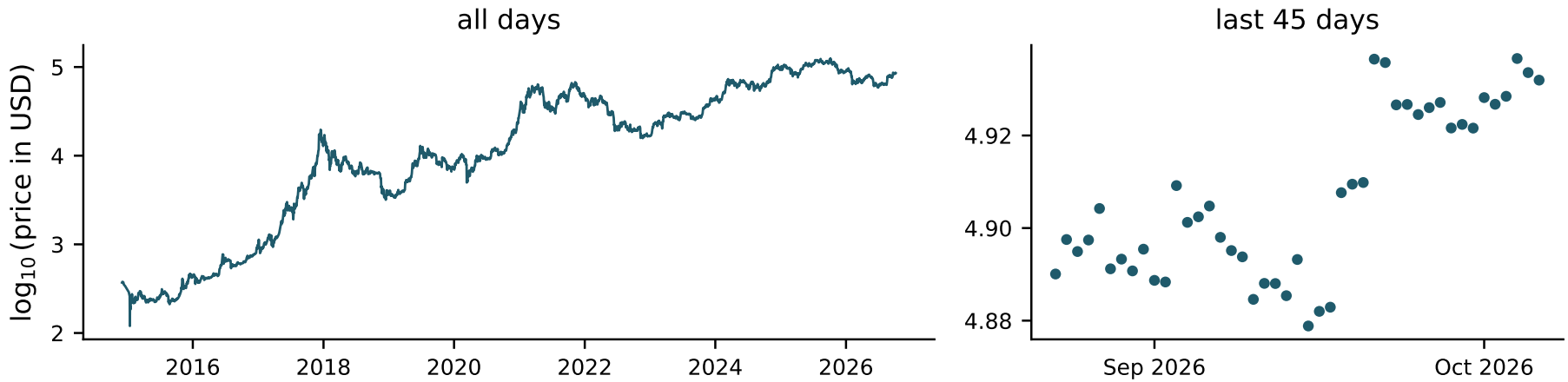
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Goal: Predict the price tomorrow, and the next day ...

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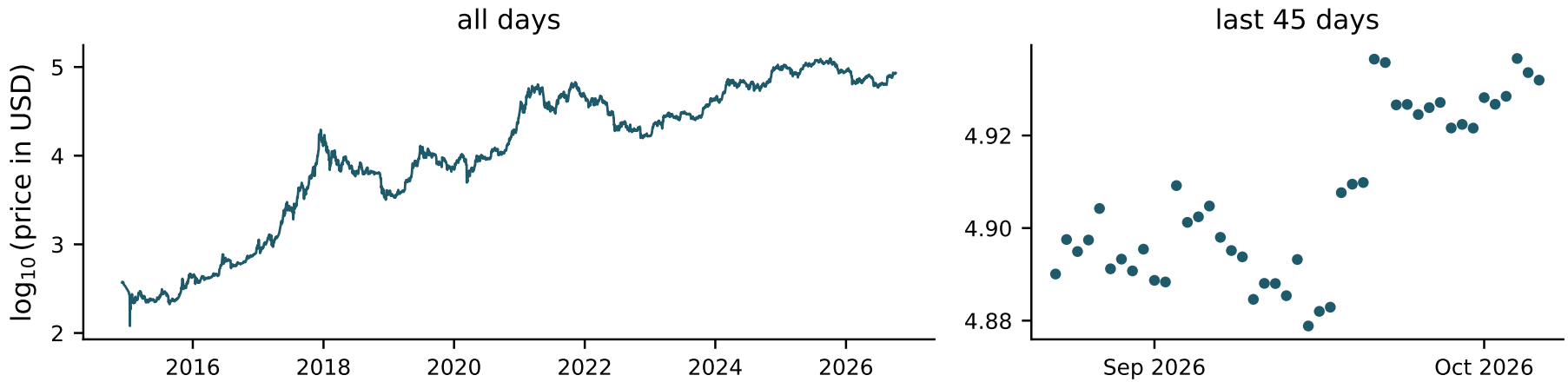


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- We could try treating time as a predictor, but this doesn't really make sense: the price on each day is not produced independently. In other words, the data points (t, Y_t) and $(t+1, Y_{t+1})$ are not independent.

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- We could try treating time as a predictor, but this doesn't really make sense: the price on each day is not produced independently. In other words, the data points (t, Y_t) and $(t+1, Y_{t+1})$ are not independent.
- Instead, we want to capture the fact that prices depend on the previous prices.

Autoregressive models

An autoregressive model (AR(1)) is a linear regression model where the predictor is the previous value of the response:

$$Y_t \mid Y_{t-1} \sim \text{Normal}(\beta_0 + \beta_1 Y_{t-1},\ \sigma_\epsilon^2), \qquad \text{i.e.} \qquad Y_t = \beta_0 + \beta_1 Y_{t-1} + \epsilon_t.$$

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The data for the regression are the pairs

$$(X_1, Y_1), (X_2, Y_2), \ldots, (X_{N-1}, Y_{N-1}) = (Y_1, Y_2),\ (Y_2, Y_3), \ldots, (Y_{N-1}, Y_N).$$

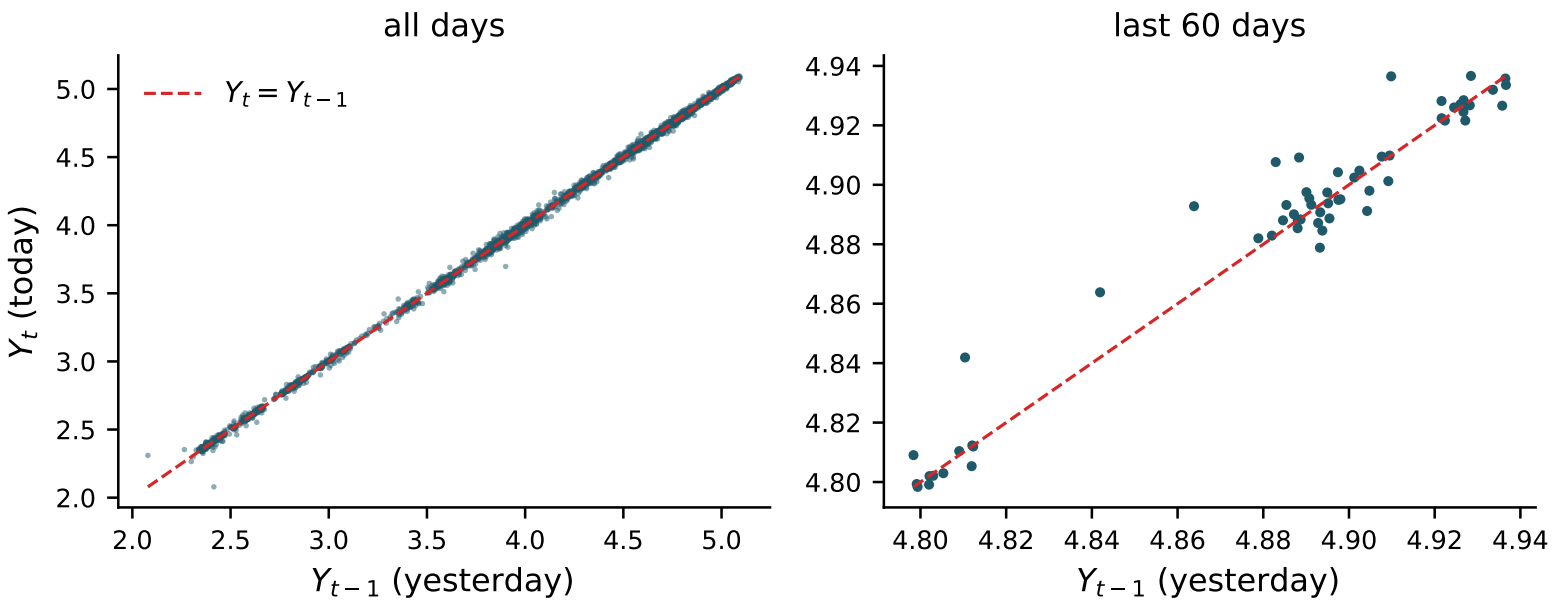
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Stationary process

- A series is stationary if Y_t and Y_{t-1} have the same mean μ and the same standard deviation σ_Y. Then

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$$\mu = \beta_0 + \rho\mu \implies \beta_0 = (1-\rho)\mu$$

- Therefore

$$\boxed{Y_t = \mu(1-\rho) + \rho Y_{t-1} + \epsilon_t.}$$

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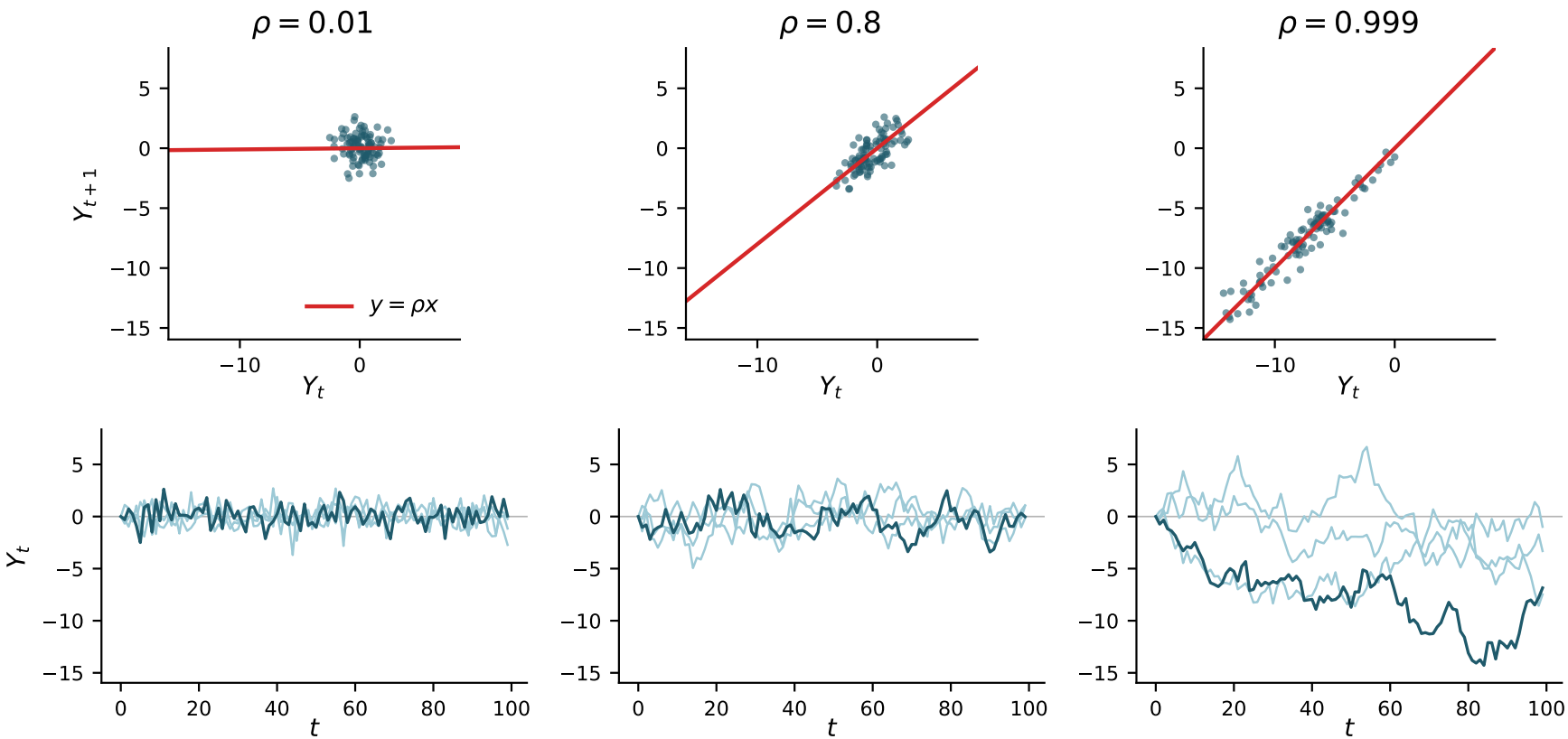
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- Taking the variance

$$\sigma_Y^2 = \sigma_Y^2\rho^2 + \sigma_\epsilon^2 \implies \sigma_Y^2 = \frac{\sigma_\epsilon^2}{1-\rho^2}$$

Simulated AR(1) models

$Y_{t+1} = \rho Y_t + \epsilon_{t+1}$ with $\epsilon \sim \text{Normal}(0, 1)$. Top: Y_{t+1} vs. Y_t for one series; bottom: a few series of 100 steps.



Estimating ρ: The Hurwicz bias

Take $\mu = 0$ for simplicity. The mean of the least squares estimator is

$$E[\hat{\rho}] = E\left[\frac{\sum_t Y_{t-1} Y_t}{\sum_t Y_{t-1}^2}\right]$$

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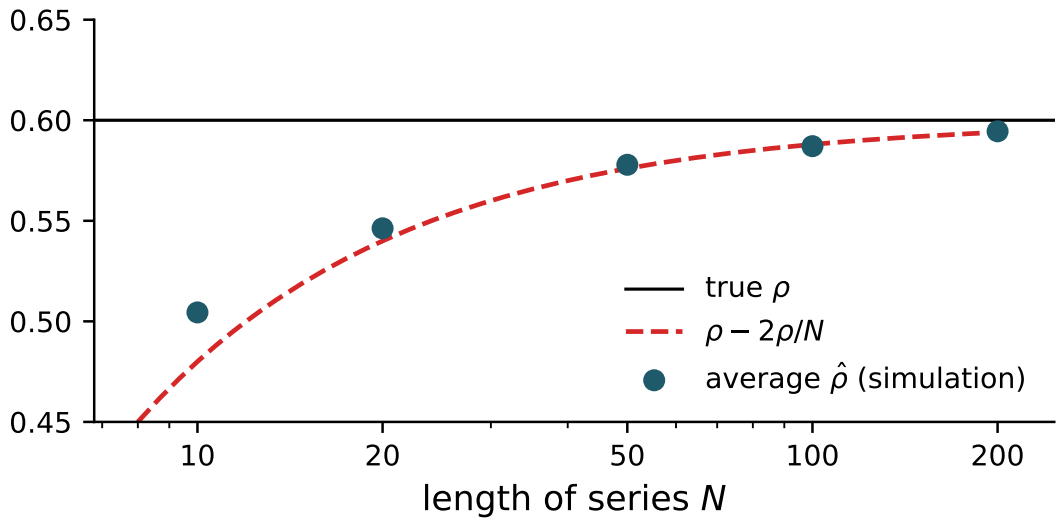
In ordinary regression, the noise ϵ_i is independent of *all* of the predictors, so conditional on X the second term has mean 0 and $\hat{\beta}_1$ is unbiased.

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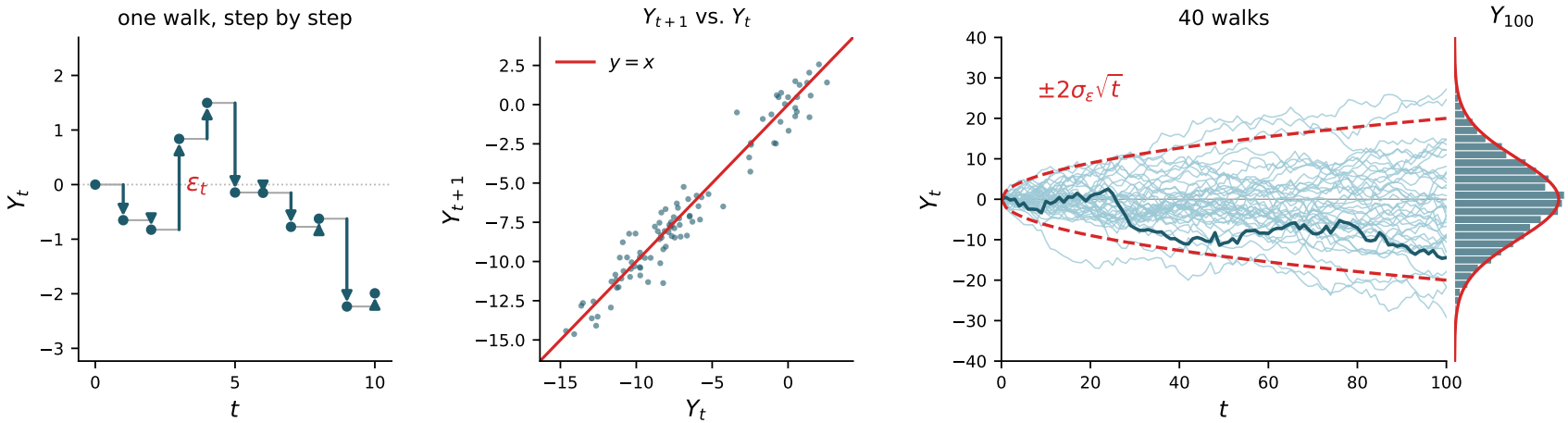
When $\rho \approx 1$: The random walk

If $\rho \approx 1$, then $\sigma_Y \approx \infty$. How can this be?

$$Y_{t+1} = Y_t + \epsilon_{t+1} = Y_{t-1} + \epsilon_t + \epsilon_{t+1} = \cdots = \sum_{s=1}^{t+1} \epsilon_s \qquad (Y_0 = 0)$$

And hence the marginal distribution of Y_t is

$$Y_t \sim \text{Normal}(0, \sigma_\epsilon^2 t)$$



How fast do correlations decay?

The autocorrelation at lag k is $\rho_k = \mathsf{corr}(Y_t, Y_{t+k})$. For a stationary AR(1) series with mean 0, $Y_{t+1} = \rho Y_t + \epsilon_{t+1}$:

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So $\mathsf{cov}(Y_t, Y_{t+k}) = \rho^k \mathsf{var}(Y_t) + 0 = \rho^k \sigma_Y^2$, and since $\mathsf{var}(Y_{t+k}) = \mathsf{var}(Y_t) = \sigma_Y^2$,

$$
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$$

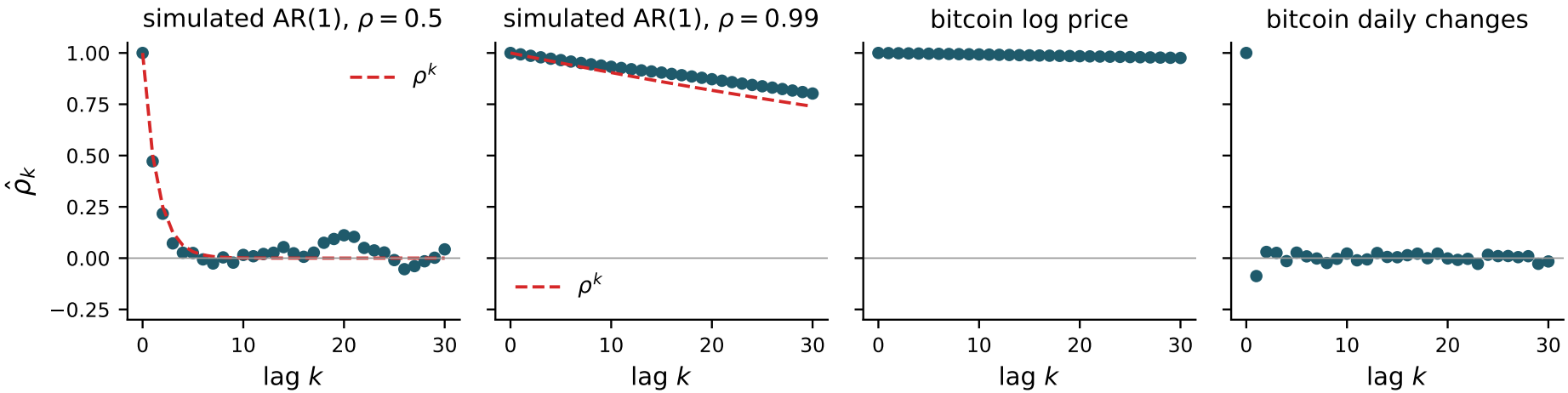
So $\mathsf{cov}(Y_t, Y_{t+k}) = \rho^k \mathsf{var}(Y_t) + 0 = \rho^k \sigma_Y^2$, and since $\mathsf{var}(Y_{t+k}) = \mathsf{var}(Y_t) = \sigma_Y^2$,

$$
\boxed{\rho_k = \frac{\mathsf{cov}(Y_t, Y_{t+k})}{\sigma_Y\,\sigma_Y} = \rho^k}
$$

The correlation decays *exponentially* in the lag: after $k = 10$ steps, $0.5^{10} \approx 0.001$, $0.9^{10} \approx 0.35$, $0.99^{10} \approx 0.90$.

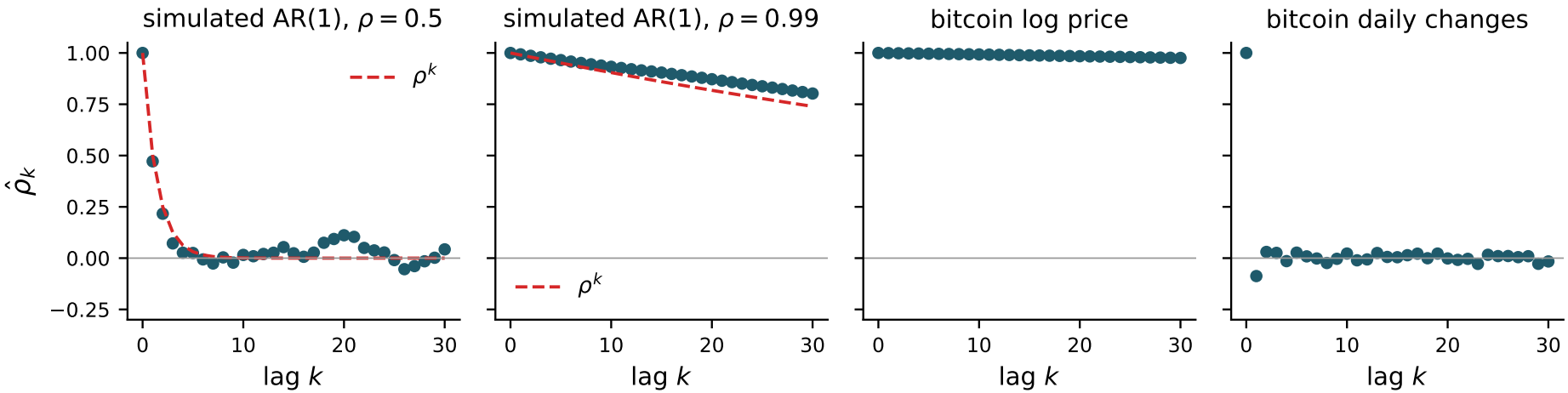
Autocorrelation functions

Sample version: $\hat{\rho}_k = \dfrac{\sum_t (Y_t - \overline{Y})(Y_{t+k} - \overline{Y})}{\sum_t (Y_t - \overline{Y})^2}$.



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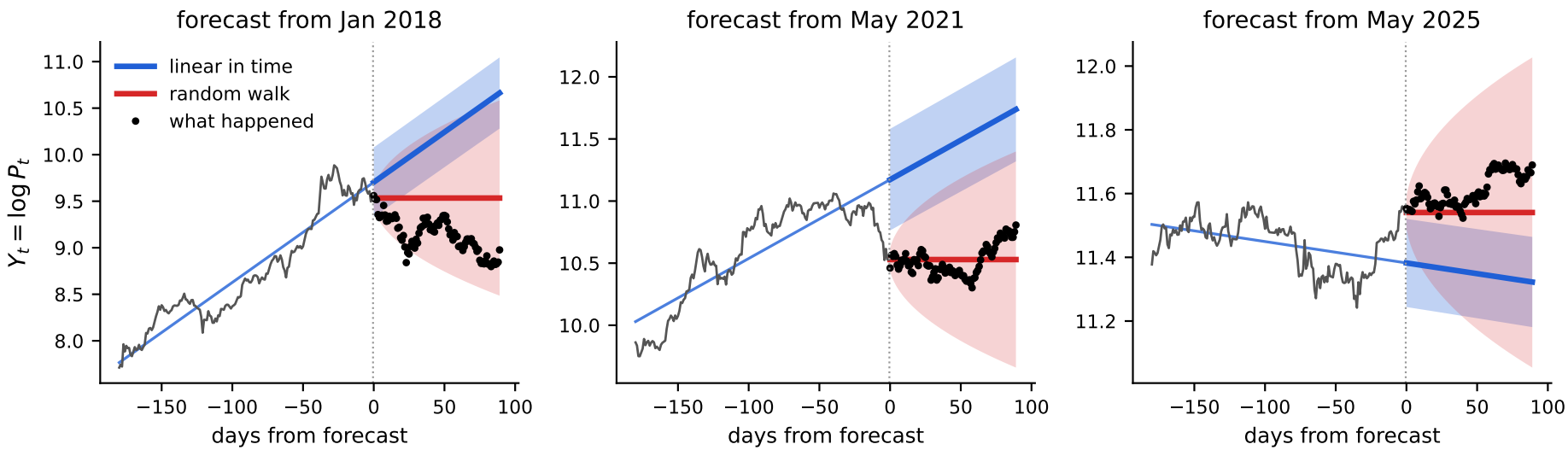


- For the simulated series, $\hat{\rho}_k$ follows ρ^k.
- The log price itself stays correlated for a long time; its daily changes $Y_t - Y_{t-1}$ are almost uncorrelated.

RW vs. linear extrapolation

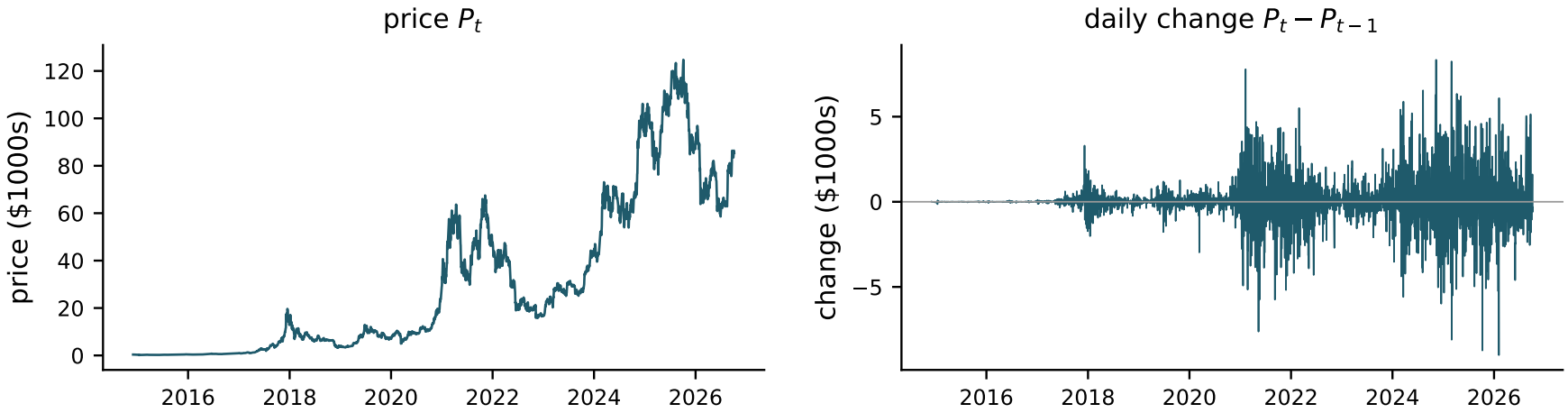
Fit two models to $N = 180$ days of log price and forecast the next 90 days:

- linear in time: Fitting $Y_t = \beta_0 + \beta_1 t + \epsilon_t$ and extrapolating. Shaded area is the 95% prediction band, including both the noise ϵ and the uncertainty in the fitted line.
- random walk: Forecast $Y_{t+h} \approx Y_t$, with 95% band $\pm 1.96\,\hat{\sigma}_\epsilon \sqrt{h}$.



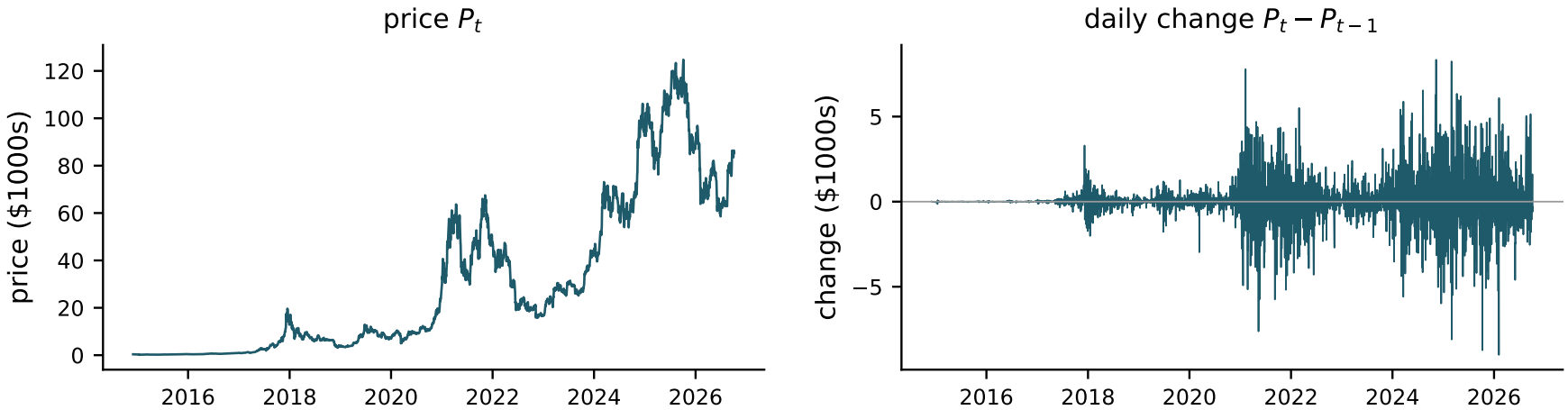
Prices vs. log prices (optional)

So far $Y_t = \log P_t$, where P_t is the price. On a linear scale:



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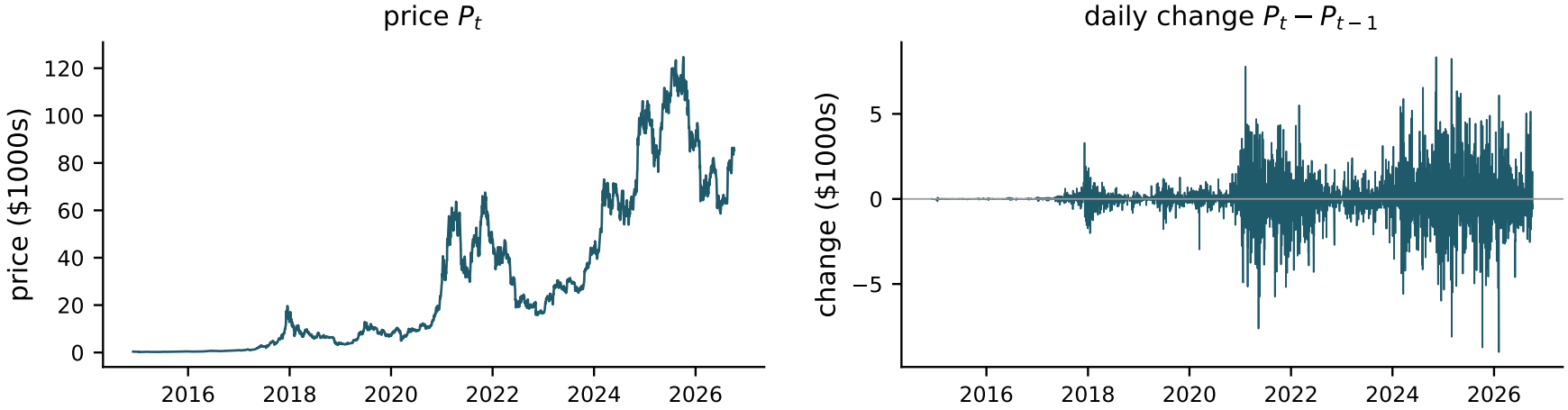
If the log price is (close to) a random walk, $Y_{t+1} = Y_t + \epsilon_{t+1}$, then

$$P_{t+1} = e^{Y_{t+1}} = P_t \, e^{\epsilon_{t+1}} \approx P_t \left(1 + \epsilon_{t+1}\right) :$$

each day the price is *multiplied* by a random factor (a multiplicative process).

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each day the price is *multiplied* by a random factor (a multiplicative process). So $P_{t+1} - P_t \approx P_t \, \epsilon_{t+1}$: the daily change is proportional to the current price. The relative volatility is constant ($\hat{\sigma}_\epsilon \approx 0.04$, about 4% a day), but in dollars the swings grow with the price: a typical daily move was about \$3 in 2015 and about \$900 in the last year.